



Laboratory work №1

Databases

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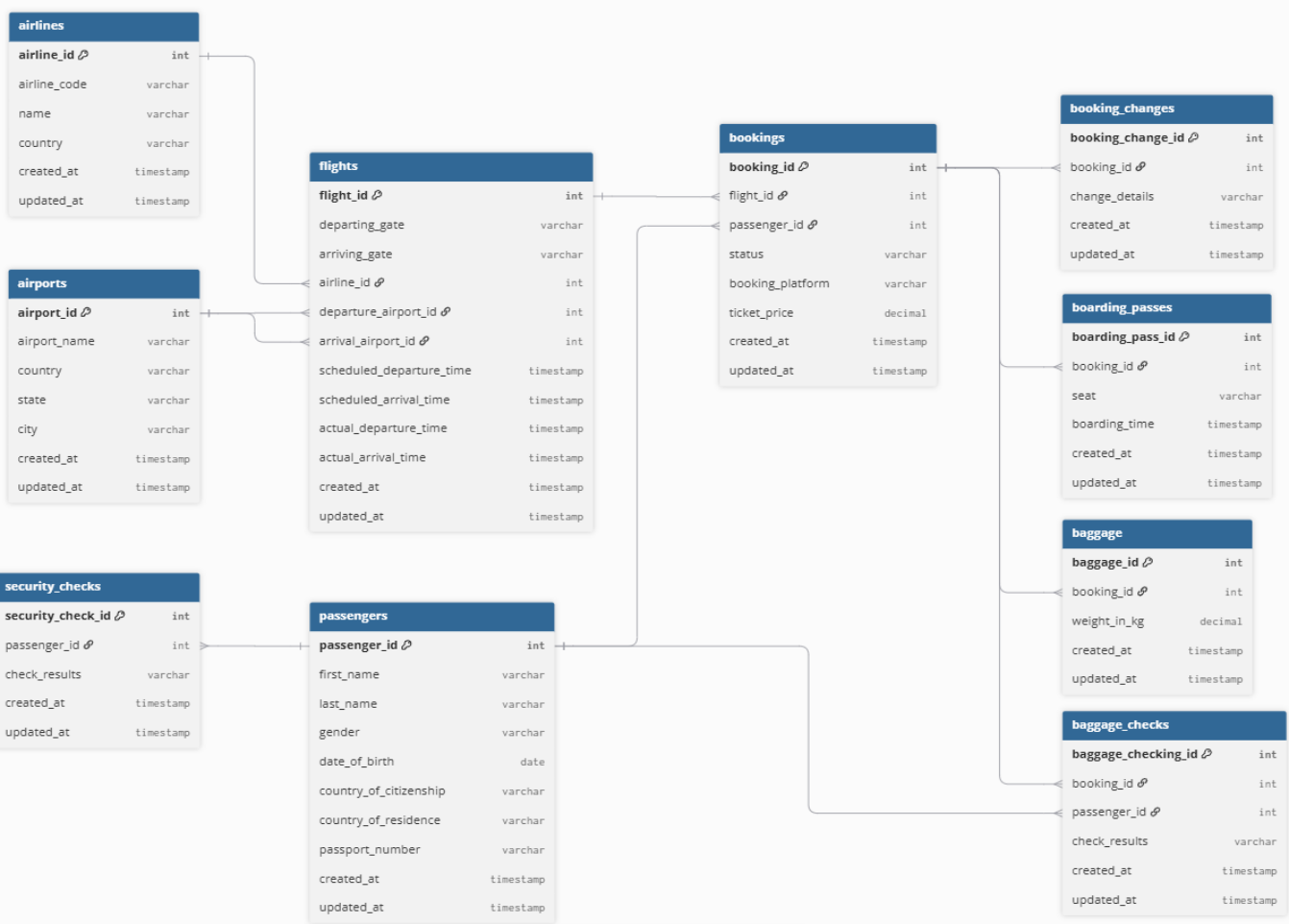
1. SYSTEM OVERVIEW & OBJECTIVES

The International Airport Management System database is designed to manage core airport operational workflows, including passenger information, flight scheduling, airline data, ticket bookings, audit histories, boarding pass issuance, baggage tracking, and security screening operations.

The primary objective of this database design is to ensure data integrity, eliminate redundancy through Third Normal Form (3NF) normalization, and maintain strict referential integrity across all system processes.

2. IDENTIFIED ENTITIES & ATTRIBUTE SPECIFICATIONS

The system comprises 10 primary entities. Below is the Data Dictionary specifying attributes, data types, and key constraints for each entity



2.1 Entity: airports

- `airport_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `airport_name` : VARCHAR(100) (NOT NULL)
- `country` : VARCHAR(50) (NOT NULL)
- `state` : VARCHAR(50) (Nullable)
- `city` : VARCHAR(50) (NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.2 Entity: airlines

- `airline_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `airline_code` : VARCHAR(10) (Unique, NOT NULL)
- `name` : VARCHAR(100) (NOT NULL)
- `country` : VARCHAR(50) (NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.3 Entity: flights

- `flight_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `departing_gate` : VARCHAR(10) (NOT NULL)
- `arriving_gate` : VARCHAR(10) (NOT NULL)
- `airline_id` : INT (Foreign Key -> airlines.airline_id, NOT NULL)
- `departure_airport_id` : INT (Foreign Key -> airports.airport_id, NOT NULL)
- `arrival_airport_id` : INT (Foreign Key -> airports.airport_id, NOT NULL)
- `scheduled_departure_time` : TIMESTAMP (NOT NULL)
- `scheduled_arrival_time` : TIMESTAMP (NOT NULL)
- `actual_departure_time` : TIMESTAMP (Nullable)
- `actual_arrival_time` : TIMESTAMP (Nullable)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.4 Entity: passengers

- `passenger_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `first_name` : VARCHAR(50) (NOT NULL)
- `last_name` : VARCHAR(50) (NOT NULL)
- `gender` : VARCHAR(10) (NOT NULL)
- `date_of_birth` : DATE (NOT NULL)
- `country_of_citizenship` : VARCHAR(50) (NOT NULL)
- `country_of_residence` : VARCHAR(50) (NOT NULL)
- `passport_number` : VARCHAR(20) (Unique, NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.5 Entity: bookings

- `booking_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `flight_id` : INT (Foreign Key -> flights.flight_id, NOT NULL)
- `passenger_id` : INT (Foreign Key -> passengers.passenger_id, NOT NULL)
- `status` : VARCHAR(20) (NOT NULL, e.g., 'Confirmed', 'Cancelled')
- `booking_platform` : VARCHAR(50) (NOT NULL, e.g., 'Web', 'Mobile App')
- `ticket_price` : DECIMAL(10,2) (NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.6 Entity: booking_changes

- `booking_change_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `booking_id` : INT (Foreign Key -> bookings.booking_id, NOT NULL)
- `change_details` : TEXT (NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.7 Entity: boarding_passes

- `boarding_pass_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `booking_id` : INT (Foreign Key -> bookings.booking_id, Unique, NOT NULL)
- `seat` : VARCHAR(10) (NOT NULL)
- `boarding_time` : TIMESTAMP (NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.8 Entity: baggage

- `baggage_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `booking_id` : INT (Foreign Key -> `bookings.booking_id`, NOT NULL)
- `weight_in_kg` : DECIMAL(5,2) (NOT NULL)
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

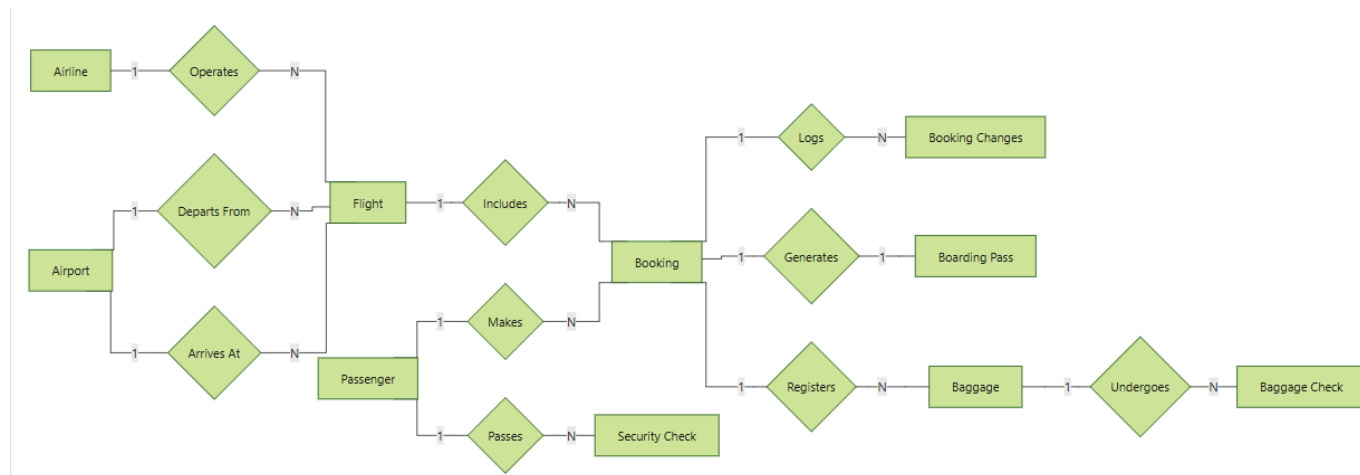
2.9 Entity: `baggage_checks`

- `baggage_checking_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `booking_id` : INT (Foreign Key -> `bookings.booking_id`, NOT NULL)
- `passenger_id` : INT (Foreign Key -> `passengers.passenger_id`, NOT NULL)
- `check_results` : VARCHAR(50) (NOT NULL, e.g., 'Cleared', 'Flagged')
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

2.10 Entity: `security_checks`

- `security_check_id` : INT (Primary Key, Auto-Increment, NOT NULL)
- `passenger_id` : INT (Foreign Key -> `passengers.passenger_id`, NOT NULL)
- `check_results` : VARCHAR(50) (NOT NULL, e.g., 'Passed', 'Failed')
- `created_at` : TIMESTAMP (NOT NULL, System Timestamp)
- `updated_at` : TIMESTAMP (NOT NULL, System Timestamp)

3. RELATIONSHIPS, CARDINALITY & PARTICIPATION CONSTRAINTS



1. `airlines` -- [`Operates`] --> `flights`

- Cardinality: 1 : N (One Airline operates Many Flights)
- Participation: Mandatory on Flight, Optional on Airline.

2. `airports` -- [`Departs From`] --> `flights`

- Cardinality: 1 : N (One Airport serves as departure for Many Flights)
- Foreign Key: `flights.departure_airport_id` -> `airports.airport_id`

3. `airports` -- [`Arrives At`] --> `flights`

- Cardinality: 1 : N (One Airport serves as arrival destination for Many Flights)
- Foreign Key: `flights.arrival_airport_id` -> `airports.airport_id`

4. `flights` -- [`Includes`] --> `bookings`

- Cardinality: 1 : N (One Flight contains Many Bookings)
- Participation: Mandatory on Booking side.

5. `passengers` -- [`Makes`] --> `bookings`

- Cardinality: 1 : N (One Passenger makes Many Bookings)
- Participation: Mandatory on Booking side.

6. `bookings` -- [`Logs`] --> `booking_changes`

- Cardinality: 1 : N (One Booking logs Many Changes over time)

- Participation: Optional on Booking, Mandatory on Booking Changes.

7. bookings -- [Generates] --> boarding_passes

- Cardinality: 1 : 1 (One Booking generates Exactly One Boarding Pass)

- Foreign Key Constraint: boarding_passes.booking_id is UNIQUE.

8. bookings -- [Registers] --> baggage

- Cardinality: 1 : N (One Booking registers Zero or Many Baggage items)

- Participation: Optional on Booking side (passengers may travel without baggage).

9. bookings / passengers -- [Undergoes] --> baggage_checks

- Cardinality: 1 : N (Baggage checks are associated with a Booking and Passenger)

10. passengers -- [Passes] --> security_checks

- Cardinality: 1 : N (One Passenger completes One or Many Security Screening Checks)

4. DIAGRAM NOTATION LEGEND & KEY

- Entity Box (Rectangle): Represents a data table storing real-world domain records.

- Primary Key (PK): Unique attribute identifier for records within an entity.

- Foreign Key (FK): Referential attribute linking child table rows to parent table rows.

- Verb Phrase / Relationship Diamond: Describes operational interactions between entities.

- Cardinality Notation:

- * 1 : 1 => One-to-One mapping.

- * 1 : N => One-to-Many mapping.